

## Original article

# “Local people want to keep their sand”: Variations in community perceptions and everyday resistance to sand mining across the Red River, Vietnam

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## ABSTRACT

The Red River Delta is Vietnam's most densely populated catchment, with its communities intertwined with, and dependent upon, the delta and its resources. Intensive, widespread riverine sand extraction is linked to substantial degradation of the delta, directly threatening these livelihoods. The response of rural communities is to resist in various ways, dependent upon a complex matrix of geospatial factors. Using an ethnographic approach, we explore factors contributing to the scale and patterns of resistance to sand mining of both an upstream riverine community and downstream coastal community in the Red River Delta. The riverine community protest openly to illegal mining directly threatening their river-connected livelihoods, manifest through reporting to authorities, bankside protests and involvement of mass media. In contrast, the coastal community have shifted from open protest, through openly reporting to the authorities, to closed forms of within-village gossip. Here, resistance patterns are driven by proximity to mining sites, perceived impact of mining, and pre-existing structural legislative mechanisms. Under scenarios of rapidly increasing rates of sand extraction to satisfy wider global needs, there is an urgent need for studies of local-scale impacts and resistance, capturing socio-economic characteristics of communities, to better plan for future scenarios of resource extraction and exploitation.

## 1. Introduction

Since the 1970s, global demand for riverine sand has consistently increased such that sand is now the world's third most utilised resource after air and water (Ardhanari, 2021; Marschke et al., 2021; Miatto et al., 2017). Torres et al. (2017) have declared a 'looming tragedy of the sand commons' due to rapidly rising issues of under-regulation and overexploitation of sand extraction. As there are currently no cheap viable alternatives to construction-grade sand derived from fluvial environments, the consumption of riverine sand is rising due to urban growth and population increase (Ardhanari, 2021). By 2030, the world's urban population is expected to grow significantly, with localised increases anticipated within urban cities in the Global South (Marschke et al., 2021). The outcome is likely an increase in localised, intensive demand for aggregates to support urban growth.

Sand is considered a common resource that is often exploited without full consideration, or exploration, of its ownership. Ownership of natural resources is often contested, with indigenous communities connected to resources in terms of their land, homes, and livelihoods. However, these claims to ownership are often neglected from resource right allocation. For example, in Myanmar, where new alluvial islands are regularly formed, mining companies are often granted rights to the land, rather than indigenous communities who traditionally have used these lands for riparian agriculture (Lamb et al., 2019). Throughout Southeast Asia, the use of fluvial sand deposits and riverbanks for subsistence, and semi-subsistence farming by indigenous communities is widespread. These livelihoods can therefore be directly affected by any alteration to deltaic systems; natural or human. Sand mining directly removes the very resource underpinning a range of livelihoods, and further contributes to pressures on other sources of livelihood through major

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destabilisation of the broader physical functioning of river and delta systems (Hackney et al., 2020, 2021; Marschke, 2017).

The direct and indirect impacts of riverine sand extraction are manifest as diminished downstream water access, altered sediment availability, riverbank instability and increased salinity intrusion (Eslami et al., 2021; Hackney et al., 2020). The associated livelihood implications often lead to conflict between users of the resource (Bisht, 2021; Franks et al., 2014). From a political ecology lens, socio-ecological resource conflicts can be considered as a struggle over the inequitable access, distribution, and control of natural resources, and the consequent social and ecological impacts (both benefits and risks; Bryant and Bailey, 1997; Pichler and Brad, 2016). Competing users of natural resources include local communities, indigenous groups, corporations, state and government, and non-governmental organisations (Neumann, 2015; Zhu, 2022), where there are competing interests and beliefs over use and ownership of sand resources. Often however, disproportionately powerful actors control the right to sand resources, excluding access and benefits from local, often marginalised, communities already vulnerable to unequal power distribution and social inequalities across gender, social class, and ethnicities (Bisht, 2021; Conde and Le Billon, 2017).

The extractive industries exert competition over natural resources, often leading to social unrest (Paredes, 2016). As resource extraction is often associated with the degradation of local livelihoods, it may lead to localised forms of social movements, protest, and resistance (Caretta and Zaragocin, 2020). Protests and resistance arise from a complex matrix of factors, ranging from noise disturbance, forced eviction from homelands without compensation, to the interruption and degradation of livelihoods with associated negative wellbeing outcomes (Banerjee et al., 2021). Principally, resistance consists of opposition to existing power relations in the form of action - verbally, cognitively, or physically (Conde and Le Billon, 2017; Hollander and Einwohner, 2004). Conflict and resistance are often the responses to natural resource extraction undermining the livelihoods of local populations reliant on the environment (Steinberg, 2016).

Documented community resistance varies from large-scale social movements to everyday, internal microlevel protest within small communities (Lassila, 2018; Purnomo et al., 2021; Scott, 1990). Types of protest to resource conflict often take the form of closed and/or open resistance, where, in the former, protesters do not appear in public, and reaction in the latter occurs publicly, with the problem, action, and protestors being clearly apparent (Purnomo et al., 2021; Scott, 1985, 1990; Wicaksana and Amirudin, 2021). For example, local communities of the Semarang region, Indonesia, resisted sand mining through hidden, internal means despite environmental degradation taking place because of, in part, employment opportunities arising (Wicaksana and Amirudin, 2021). A further form of resistance is through existing culture and habits of local communities, operating as cultural resistance (Purnomo et al., 2021). This type of resistance can be mediated by cultural norms and practices from broad scale to micro-level practices and traditions within a community. Options for resistance are also often led by the pre-existing structural, legislative mechanisms in place (Conde and Le Billon, 2017). The transformation of local power relations through companies and governmental bodies renting out community lands - often supported through legislation permitting exploration, exploration permits, and natural resource extraction - significantly shifts wealth distribution and land ownership away from resource-dependent local people, who as a result are often excluded from local decision-making (Rasch and Köhne, 2016). This can result in structural forms of resistance, with protest forms following bureaucracy, policy, and governors at all scales (Purnomo et al., 2021).

In the Mekong Delta, communities did not complain to the court about a sand mining operation as they '*don't think we could win against very rich and powerful people*' (Rousseau and Marschke, 2023, p.6), graphically illustrating the role of power dynamics in community protests. In some cases, political regimes can exert control over the type and scale of protest; for example, the Cambodian government have notably

detained and arrested environmental activists protesting sand mining (Len, 2017; Van Arragon, 2021). This high-level intervention may incite fear in protesters, resulting in more closed and hidden anecdotal protest. Yet, not all communities resist, and do not always protest with the same objectives, vigour, or timeframe (Conde and Le Billon, 2007). The form, and level, of resistance taken is highly complex and influenced by conditions such as legislative processes, culture, the extent and scale of resource extraction, perceived resource rights, and the level of impact to livelihoods and ecosystems. However, this is poorly understood in the context of micro-level livelihoods i.e., at household and community level, meaning there is little focus on community resistance to the extractive industries and the resource conflicts they engender.

Local communities are rarely considered in mitigation against anthropogenic disturbance; they receive no compensation for losses resulting from natural hazards enhanced by extraction of resources, yet are the first to experience the impacts to their land and livelihoods. It is therefore unsurprising that the focus of national or regional policy has not been on the impacts on local livelihoods (Lamb et al., 2019). In general, the social impacts of aggregate mining are understudied, and very little is understood about the livelihood vulnerabilities produced by sand extraction. In addition, there has been little research into the interaction of sand mining, livelihoods, and human-induced ecosystem change (Koehnken et al., 2020; Lamb et al., 2019).

This study aims to explore the factors contributing to the variability in the scale and patterns of resistance to riverine and coastal sand mining across the Red River Delta in northern Vietnam through in-depth study of two representative communities. Applying ethnographic techniques through the lens of political ecology, we assess the role of perceived impact of sand mining, proximity to mining site, pre-existing resistance structures and cultures, and legality of mining and use a biophysical, social, economic, and political assessment to assess the capacity of villagers to express resistance towards sand mining. In doing so, a variety of resistance patterns are revealed between the two communities, driven by extrinsic structural and political mechanisms and level of threat to livelihoods. It is critical to involve indigenous communities and delta dwellers in livelihood research, as these are the *in-situ* experts who 'know the river' and who will experience these issues differently depending upon their livelihoods. Localised knowledge is essential to contextualise livelihoods dependent upon riverbanks and riparian land and raise understanding of how peoples' lives are being impacted by sand mining and resource extraction. Capturing patterns and behaviours of resistance of communities impacted by anthropogenic activities is needed where understanding and incorporation of local viewpoints is lacking.

### 1.1. Study area and socio-economic setting

The Red River (Sông Hồng) is the second largest river in Vietnam, and plays an extremely important role in the economic, cultural, and political life of Vietnamese people (Dang et al., 2010). Economic reforms in Vietnam in the 1980s promoted deregulation and foreign investment, leading to rapid growth in agriculture, industry, construction, and housing development in the Red River catchment, particularly on the delta in and around Hanoi (Leducq and Scarwell, 2018). Between 2007 and 2018, Vietnam experienced an average population growth rate of 1.1%, with this trend almost doubling in urban areas such as Hanoi, and is on the rise (Novellino et al., 2021). Urbanisation and associated development to support population growth requires substantial volumes of sand and aggregates to fulfil intensifying demands for construction materials for urban development. It is likely that these trends are only set to increase in the future as urbanisation and national economic development levels increase. Throughout the Red River Delta significant erosion has been observed with villagers reporting riverbed subsidence of up to 2–3 m in Hanoi, and up to 8–10 m elsewhere (Chinh, 2018; Dinh, 2020; Phuong, 2019). There are also reports of sudden changes in the mechanism and rate of riverbank failures, where typical slow

recession has been replaced by high magnitude collapses, attributed to sand dredging undercutting the banks (Dung, 2011). This threatens local Red River communities, through alteration and loss of in-channel sand banks and the local hydrodynamics of the river network, affecting resource-dependent livelihoods.

To assess variations in the factors driving resistance, two rural villages, one riverine and one coastal, adjacent to the Red River or its distributaries, were selected to characterise and compare the impacts and response of communities to sand mining across the Red River catchment (Fig. 1). Both villages are of a similar relative spatial area and population size (Table 1). To protect the residents of the villages, located within the Vĩnh Phúc and Thái Bình Provinces, correspondent names have been anonymised, and henceforth we refer to the province names to differentiate between villages.

At both villages, sand mining has been ongoing intensively for the last 5 years, with riverine sand mining near the village in Vĩnh Phúc, and coastal mining occurring around 6 km from the village centre in Thái Bình. Both communities report a recent shift from small-scale artisanal removal using baskets, to large-scale intensive pumping of sand. At the Vĩnh Phúc village, communities report that large dredgers, capable of pumping sand from depths of 40–50 m, have exhausted sand reserves in neighbouring villages with the most significant phase of mining occurring between 2020 and 2021. Although sand mining is temporarily banned in Vĩnh Phúc Province (Decision 713/QĐ-UB), mining still occurs. The river acts as the internal political border between Vĩnh Phúc and Hanoi Provinces, meaning different provincial-level legislation applies to either side of the border. As a consequence, mining operations can occur legally on the Hanoi-side of the river, making regulation of sand mining activities difficult. For Thái Bình, despite sand mining occurring offshore, the community are aware of the activity, with a consensus view that it has been occurring since 2015. The mine has been licensed to four companies in the Hóa River estuary (Decision 1594/QĐ), a distributary channel of the Luộc River.

The upstream village in Vĩnh Phúc Province is directly adjacent to the Red River and is subsequently exposed to riverbank erosion and land

**Table 1**

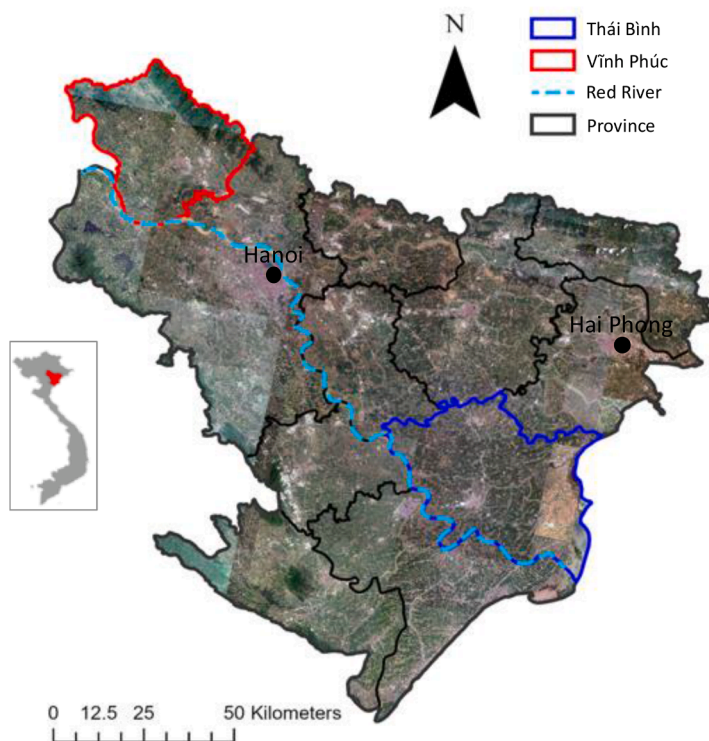
Provincial statistics for Vĩnh Phúc and Thái Bình Provinces from 2021 (CEIC, 2021; GSO, 2022).

|                                              | Vĩnh Phúc     | Thái Bình     |
|----------------------------------------------|---------------|---------------|
| Population density (person/km <sup>2</sup> ) | 964,000       | 1184,000      |
| Area (km <sup>2</sup> )                      | 1236,0        | 1584,6        |
| Monthly average income (thousand VND)        | 2567 (~\$109) | 2577 (~\$110) |
| Poverty rate (%)                             | 2.3           | 1.9           |

loss. For most village residents, dairy farming is their main livelihood occupation, and they transitioned from arable farming to rearing cows around 20 years ago. Farmland, riverbanks, and sand bars adjacent to the Red River are used to grow elephant grass (*Cenchrus purpureus*) to feed cattle. More recently, younger members of the village have moved from dairy farming to work at nearby companies for their primary income. In Thái Bình Province on the delta front of the Red River, the coastal village surveyed borders a distributary channel. The villagers depend on both freshwater and saltwater for their livelihoods, which predominantly consists of arable farming. Vegetables and cash crops comprise their main source of income, together with some freshwater and brackish aquaculture, as well as subsistence farming, seafood collecting in mangroves and small-scale business. As with the Vĩnh Phúc village, the majority of young people travel outside of the village in seek of employment opportunities.

## 1.2. Data collection and analysis

To engage effectively with members of the villages in Vĩnh Phúc and Thái Bình, six focus groups and sixteen key informant interviews (KIIs) were undertaken from July to August 2022 with the assistance of field assistants local to the Red River Delta. In focus groups, participatory methods included both group discussion and participatory mapping, with the average number of participants per focus group being nine. KIIs involved discussions with key village-level and commune-level



**Fig. 1.** Location of the northernmost (Vĩnh Phúc Province) and coastal (Thái Bình Province) study sites and main channel of the Red River. The precise location of the villages has been excluded for the sake of anonymity. Photos of sand mining vessels were taken in July 2022 by the primary author.



(Vietnam's lowest local administrative sublevel, below district) leaders and other local authority figures, who act as representatives for different collectives including the Women's Union, Youth Union, Fatherland Front and Elderly Association, who together represent the views of numerous residents of the villages and communes in both locations. The participatory research methods included three different focus groups in each village, consisting of a mixed-gender group, a women-only group, and a youth group of mixed gender comprising participants aged 13–35; the aim in all activities was to gain an in-depth, consensus view within each community, and to enable all members of the community to speak openly and freely.

For both focus groups and KIIs, a semi-structured, ethnographic interview approach was used, with themes and question prompts focussing broadly on the current livelihoods of the village, how livelihoods have changed over time, and the role of human activities on morphological impacts to their delta. The purpose of this was to understand the link between the livelihoods of local people and the Red River and its delta, as well as the factors influencing the change in livelihoods over time. Narratives and quotes were analysed using a deductive thematic framework, allowing the discourse and key themes to emerge directly from the data, as opposed to being pre-informed (Fereday and Cochrane, 2006).

## 2. Drivers and forms of resistance to sand mining

Here, we discuss the main resistance patterns exhibited by village members and key factors influencing the resistance to sand mining, before synthesising this new locally-rooted knowledge base to illustrate the relationship between the forms of resistance documents and the driving factors of resistance.

### 2.1. Drivers of resistance

For the residents of Vĩnh Phúc, who are adjacent to the Red River, sand mining directly and visibly threatens their housing and farmland on the sand banks and riverbank, which is used for subsistence and feeding livestock. Displacement, relocation and inadequate or lack of compensation threaten the subsistence of local people, particularly where there is overlap between mineral resources and local people's infrastructure and livelihoods (see ideas in Chindo, 2011; Conde and Le Billon, 2017). Community members expressed concern over loss of their homes and infrastructure, fearing relocation without compensation – a situation which had occurred in the past.

*'Tomorrow, you go to near my house right away, cracked from under the ground. When it sucked the sand and lost its legs [part of the house*

*foundation], the difference was large, the soil kept drifting out, gradually losing its legs more and more' (KII4).*

Erosion and landslides are a critical problem for the villagers in Vĩnh Phúc, whose houses and livelihoods are connected to the riverbank; the development of mining equipment has been attributed to this erosion. One focus group estimated the embankment was being undercut by 10–20 m because of sand mining. Riverbank instability and erosion is causing implications for community members' houses, where cracks on the walls of housing have appeared in both old and new housing, with homeowners stating it is potentially due to sand mining. This is shown in Fig. 2, where large cracks in both houses and the dike can clearly be seen.

The increased instability of the river system impedes on the access rights of the Thái Bình community, who perceive they have the right to use the river and its resources for leisure and secondary livelihood activities.

*'No one goes to the erosion area because it is dangerous. The landslides are dangerous, the sand is not stable. It is slowly falling into the sea day by day' (FG4).*

In these cases, communities' patterns of resistance to mining were seen to comprise open, public protest. As the Thái Bình village is located around 6 km from the mining site, mining activity is not immediately visible to the community members, especially as they have no boats or other means by which to visit the estuarine sand mining site. Mangroves are relied upon as a secondary source of income to the villagers' arable farming livelihoods, and its importance to the community is very high, given it offers a range of protective and livelihood sources.

*'There are so many benefits of the mangrove forest from when it started growing. It prevents the dike from breaking, the air is fresh, it supplies food for the local people, it keeps the land' (KII12).*

The spatial dislocation between the community and sand mining activity means that resistance patterns may be more closed and internalised, i.e., not going beyond the village. Despite this, the impacts of sand mining on the willingness of the Thái Bình community to access the livelihoods provided by the mangrove forest are keenly felt.

*'Before, adults go to here often, go to the mangrove to collect seafood, and to the sea. Since 7 years ago [sic], we don't go here because it is dangerous. Our parents forbid us to go, we are afraid if water comes, we will drown' (FG4).*

It is therefore likely that increased visibility of the impacts of resource extraction enhance the likelihood of community mobilisation (Conde and Kallis, 2012) against such activity. One focus group, who when asked about their hopes for the future, stated they wanted:



**Fig. 2.** Cracks in a house (shown in a VTV1 broadcast) and the cracked embankment in the Vĩnh Phúc village (photo taken August 2022). Solid lines show the extent of the vegetated embankment adjacent to the river, with dashed lines showing damage to the concrete dike running parallel to the vegetated embankment.

*'No sand mining, don't want it to cause holes, change the currents, we want to feel safe to go to the mudflat'* (FG5).

Local people are very cognisant of the numerous ways sand extraction is directly impeding their lives, spanning from livelihoods to their wellbeing; this acute awareness of extraction-related impacts may contribute to the level of protest.

A common theme in both communities identified as a driver of resistance is the connection between sand mining operations and the people undertaking the extraction. In Thái Bình, local people have identified that no money from the licensed Hôa River estuary mine has come to the community, as the mine is owned by people outside of their village, stating:

*'We hate sand mining. The income goes to the Province, it does not go to us'* (KII11).

Villagers in Thái Bình are aware that although the sand mine is within their commune, the mining license does not entitle local people or the commune to funds from the operation, and the benefits end up outside of the village. In Thái Bình, the sand mine is in proximity to shrimp, fish, clam, and bivalve farming, so although mining directly conflicts with these livelihoods, the majority of fishermen are not local to the village. There is an understanding amongst villagers that the whole fish farming area and sand mining operation is external to the village, which *'belongs to Thái Bình'* (KII17). This could result in relatively lower level of resistance; although the mangrove property rights are open access, the farm and mine is clearly owned and operated by companies external to the village. When a sand mine is granted by the Provincial People's Committee and the Department of Natural Resources and Environment the commune must inform its local communities, making it immediately clear to local people that the land is no longer accessible. Resource rights are removed from the villagers, as the Thái Bình community are not permitted to access or use the coastal sand resource at the mining site, given it is operated by an external company. This is also experienced in Vĩnh Phúc where local people recognise that the sand miners are not from either their village or commune:

*'Sand mining in the river and sea, there are big boats. The people are not from here, they are from outside. It is very hard to control'* (KII13).

Conflict between communities and stakeholders from areas external to the community is common, as local people exert their right to defend their land and resources exploited by outsiders (Yasmi et al., 2010). Communities recognise their right to their natural resources and are unhappy that, as sand miners are not from their village, they receive no benefit. In Vĩnh Phúc, local people believe the sand in the Red River adjacent to their homes belongs to their village, and seek to protect the resource:

*'Local people want to keep their sand. Sand is a natural resource. The local people protect the asset themselves, for their houses, they spend money, in their whole life can only build one house. We just stand on the riverbank, if the local people get angry, they bang. Local people have no right to do anything'* (KII11).

Power imbalances are often demonstrated in resource conflicts, whereby a few individuals profit from sites of natural resources, as most local people do not have financial means or assets to partake in resource extraction (Bari and Haque, 2022; Zhu, 2020). Current practices and policy mean that local people do not have control, or the ability, to mitigate mining operations on their river. If communities lose their houses to the river, they *'have to pay out of pocket'* (FG2), which can cost upwards of 2 billion VND (~\$84,000). As a result, less funds are available to spend on healthcare, education, livelihood investments and adaptation practices. This occurred in 1997 in the Vĩnh Phúc village, where riverbank landslides resulted in houses collapsing, and the affected village members received no compensation. The communities do not know if their homes are at risk of subsidence if sand mining

continues; instead, they are increasingly aware as to their lack of resource rights. The outcome is communities recognising that sand is extracted by sand miners who are "outsiders" to their village, and that consequently they have no right to the resource. Resistance is driven by feelings of injustice; the overwhelming and unequitable impacts to their livelihoods, compounded by the lack of compensation from sand mining within the confines of their homes. The result is that the communities become increasingly marginalised, losing their sense of voice within the resource conflict. To increase the resilience of these communities, improved rights of local people are a key step in regulating the sand industry (UNEP, 2022).

## 2.2. Forms of resistance

### 2.2.1. Reporting

Reporting of resource or land-use issues by local communities to local-level authorities is a historical form of protest in Vietnam (Le, 2020). For example, historical resistance to the *đổi mới* reform in Thái Bình Province took the form of organised complaints, the majority of which were made by Catholic women, to district and provincial levels to listen to their issues, which led to the end of heavy agricultural taxes (Kerkvliet, 1995; Le, 2020). In Thái Bình, community representatives made complaints to the commune-level authority about the sand mining operation. One community member explained that the local authority meets with the representative of different organisations, such as the Women's Union, every year to discuss issues within the village. They stated they complained to the commune authority at one of these meetings, where *'5 years ago, [we] resisted sand mining. Gave complaint to the commune authority'* (KII12). However, following this meeting they were informed that the mine was legal and licensed by the government:

*'Since 3 years [sic] – the government issue the licence, and the local people don't complain because they know it is no longer illegal, it is legal now. They always complained before, the government say it is to build the road'* (KII12).

Subsequently, once the village representative was made aware by local government that sand mining was legal, they felt they could no longer complain about the mine, because legislation was in place to license the coastal mining area (Decision 1594/QĐ). In February 2020, Thái Bình's Provincial People's Committee granted 6 mining licenses in the Hôa River from 2020 until 2025, and 7 new mines from 2026 to 2030, with a total forecasted resource of 29,357,000m<sup>3</sup> and 33,378,400m<sup>3</sup> for each year respectively (Decision 03/2020/QĐ-UBND). New mines have been granted to serve the coastal road project linking Quảng Ninh, Hải Phòng, Nam Định and Thái Bình Provinces. As of February 2020, mining licenses in Vietnam are granted by the Provincial People's Committee and the Department of Natural Resources and Environment (Decree 23/2020/NĐ-CP). Given that the mines are licensed by higher levels in government, and no money can legally go to the village, local people do not protest to authority, and only make internalised complaints amongst community members. This is a demonstration of the transition to open structural to closed, or hidden, cultural protest, where community members no longer complain freely, keeping their resistance within their village, and to the village head (Purnomo et al., 2021; Scott, 1985).

Villagers are no longer given an official structural platform to complain openly about sand mining, as they have been informed the mining site is legal and that there is legislation to support this (Decision 03/2020/QĐ-UBND). So, protest is only permitted within the village, and not beyond the commune-level authority. Local people have no choice other than to accept the decisions and policies in place, removing their right to protest sand mining. Further, continual protest requires time and resources, often not available to communities reliant on support from secondary livelihood activities. This lessens their ability to openly protest (Wicaksana and Amirudin, 2021). Despite the village being open about the extent of environmental degradation caused by the

mining operation, the community are not entirely dependent upon open-access resources from the coastal area, meaning their primary livelihood activities are not immediately threatened. Furthermore, there is also an understanding that sand mining is occurring throughout Vietnam, where a participant stated *'it [sand mining] happens in the whole country, not just in this area'* (FG6). This highlights that local people recognise sand mining is a widespread issue, which offers another explanation for the closed resistance pattern of Thái Bình. Feelings of helplessness can discourage open protest (Mannarini et al., 2009), so where local people feel that the issue is insignificant due to its widespread occurrence, collective open action does not occur. However, despite communities not having the ability to resist in an exposed and open manner, the situation illustrates the many frustrations experienced within the village. Although there is little opportunity for this hidden resistance to foster awareness and progression of local people's livelihood security, reduced visibility of the protest does not mean communities are not impacted (Scott, 1985).

In contrast in Vĩnh Phúc, at the point local people knew mining was unlicensed and illegal, they began to report and protest sand mining vessels: *'2019 was the first time we reported, this is the first time we knew it was illegal, before we thought it was legal'* (KII4). This is reflective of the nature of protest in northern Vietnam - if there is no legislation from the higher government to support sand mining, they can openly complain and report sand mining. There is an official procedure in place for reporting illegal mining in Vĩnh Phúc:

*'People call the village head, the village calls the commune, the commune calls the district, the district calls the river police, then the boat goes away. In 2020, once a day, the village head calls the commune police chief, and the People's Committee calls the chairman of the People's Committee. At the end of 2020, there were too many onshore, the district police came back to take pictures once'* (KII2).

As of 2019, the People's Committee of Vĩnh Phúc Province gave responsibility to villages and communes to promptly report incidences of illegal sand mining to the district on a 24/7 hotline, to enable the prompt *'clearance and handling'* of mining operations (Decision 713QĐ-UBND). Collective action and reporting to the commune have proven successful in reducing mining vessels affecting Vĩnh Phúc, with continual open structural protest complying with the official reporting mechanism in place. The sea police and district-level authorities and above are the only ones who hold power to prevent mining boats, which is widely recognised by the community members, who state *'the commune police do not have the authority to arrest them'* (KII1). As with the commune, local people feel they do not have the ability to respond themselves if illegal sand mining occurs their village, resulting in village members following structural reporting mechanisms to protest sand mining. Many participants also expressed they do not have access to boats or any means and rights to reach the sand miners directly. This is recognised in legislation, which states even most management agencies below district-level do not have means of transportation on the river (Decision 713QĐ-UBND), let alone at village-level. Local people make complaints to the commune because this is their only official structural mechanism in place for reporting and protesting illegal mining. However, reporting is not undertaken every time a mining vessel is present:

*'It is hard work, local people call if they see illegal sand mining, they notify the commune, the commune notifies the sea [waterway] police. You have to pass many steps, and the sand mining owner disappears already. Now, it is simple, we just phone directly to other people, in the district level'* (KII1).

This reflects the complexities in reporting illegal mining vessels, as well as the amount of time invested into sand mining by local people. In resource conflicts, local communities often do not have time or capacity to engage in discussions (Bebbington, 2010). Furthermore, when villagers report mining, they are asked to provide evidence of the boats; however, this is not always possible: *'the police say they need evidence. My*

*phone is bad quality, I can't take photos at night'* (KII1). Despite the requirement of villages to report illegal mining, local people cannot always provide evidence, or report to the commune and waterway police, due to lack of resources and time to invest in mitigating sand mining. Communities have little choice but to protest through the pre-existing structural mechanisms presented to them (Conde and Le Billon, 2017). Continual reporting of mining operations by village members is met with little change in sand mining operations, leading to further forms of protest. Another form of reporting is through petitions: a petition made in 2016 and sent to the commune and district contained almost all the villagers' signatures.

*'Put the document [petition] to the village, the village sent it to the commune, and the commune sent it to the district. The district replied, because this river has [borders] Hanoi and Vĩnh Phúc. In Hanoi, they allow the boats to mine for the flow of water. They [sand mining] stopped, several months after, they came back to do it at night'* (FG1).

Both petitioning and reporting are both forms of open structural resistance, whereby forms of protest aim to influence local policy-structures and engage with government leaders (Purnomo et al., 2021). The community members follow the structures presented to them to report and prevent illegal mining; however, these are not totally effective in reducing the incidences of illegal sand extraction in Vĩnh Phúc, leading to further forms of protest.

#### 2.2.2. Riverside protests

A further form of resistance to sand mining taken by local people is through directly protesting sand mining vessels when they appear in the river. This practice began in 2014, with the highest number of protests by the local community reported to occur from 2014 to 2015 and 2020 to 2021. When sand mining vessels come close to the riverbank, local people begin to protest, including shouting and making noise on the riverbank by banging pots, shining lights, and throwing rocks at mining vessels.

*'40–50 people in my village, fighting, banging gongs, the boat was also scared and left, but after a while, it came again. In general, the entire population of people near the river's edge came out to protest'* (KII2).

This is a form of open, cultural resistance, which is indicative of a response to a lack of representation in decision-making processes in sand mining operations (Ballard and Banks, 2003; Conde and Le Billon, 2017). Protests occur temporally alongside active reporting where local people feel they have a responsibility to collectively protest mining, as it is their land, village, dike, and riverbank. Simply reporting mining is not effective in reducing sand mining vessels. This also highlights distrust in the current processes in place for managing sand mining, resulting in the entirety of the village protesting on the riverbank. Formal participatory procedures often neglect the participation, perspectives, and priorities of local communities, and are therefore deemed insufficient (Özkaynak and Rodríguez-Labajos, 2017). As a result, the community state the fate of sand mining and the village will depend on the local community's actions:

*'If local people don't go against, they will do it [sand mining]. If local people don't go against them, in only 5 years, all houses will collapse'* (KII1).

Because sand mining threatens the livelihoods of local people, it has a direct impact on their riverbank properties and removes the resources the community feels belongs to them. Local people believe their only way to prevent mining is to demonstrate in riverside protests, as there are few other means to prevent illegal sand mining, heightening the emotional charge and perceived proximity of the threat. The threat of displacement is clearly a major driver in patterns of resistance and highlights the disproportionate impact of sand mining operations on local communities with existing vulnerabilities and livelihood dependencies on the environment (Conde and Le Billon, 2017; Özkaynak



and Rodriguez-Labajos, 2017).

### 2.2.3. Media

Another approach taken by the community to prevent sand mining was to contact a national television news outlet in 2020. Protesting using mass media is another form of protest by society, where social media petitions are often used to share viewpoints on illegal sand mining (Zhu, 2020). A news report showed vessels extracting sand close to the riverbank of Vinh Phúc to the whole country at 06:30am, which was reported to be effective in reducing mining by the community:

*'After the [national TV] show on the news, they stopped for several months. Even the programme shown on the TV, they stopped for 1 to 1.5 months, but it is not like everyone thinks, they still did sand mining' (FG3).*

This open cultural resistance goes beyond structural protest, where the enhanced proximity of threat results in additional forms of protest. Contacting informal networks through connections with journalists, as opposed to local authorities, enables local issues to be heard and acted upon (Karlsson et al., 2015). This media coverage of mining may act to facilitate, support, and strengthen the active protests. Although it was not effective in fully eradicating sand mining, it appears sand mining was reduced for a few months outside Vinh Phúc by open cultural protest. However, despite involving the mass media in 2020, sand mining continued.

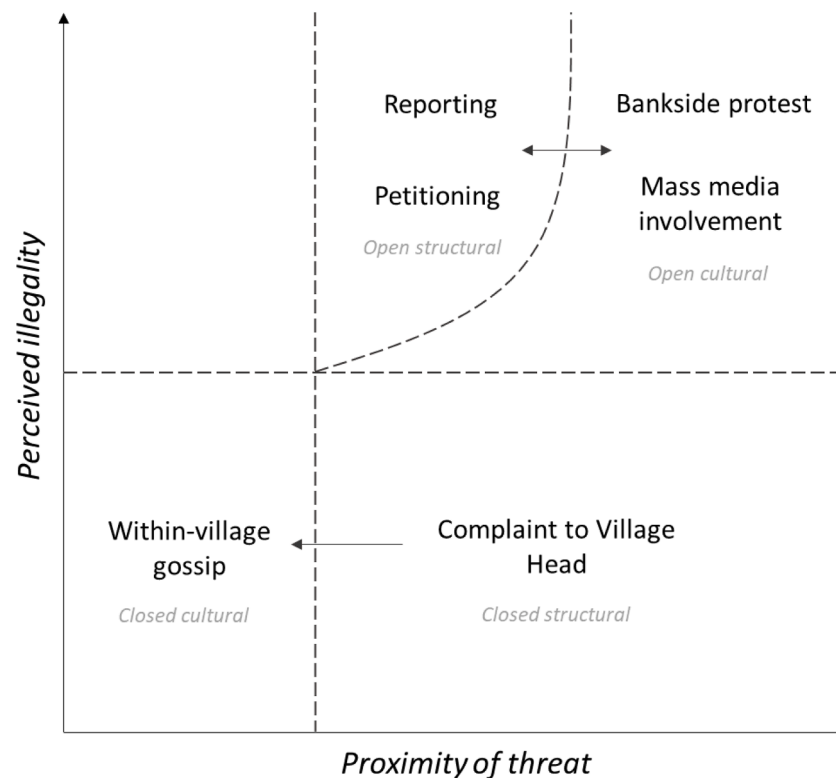
## 3. Discussion

We observe that sand mining does not directly impact the main, resource-dependent livelihood activities of both communities, but indirectly threatens them through potential loss of housing, protective dike and infrastructure in Vinh Phúc, and reduced area of mangrove offering protection of farmland and aquaculture ponds in Thái Bình. Both communities investigated exhibited similar change in their

secondary livelihood or leisure activities as a direct result of morphological change linked to sand mining activity. In Vinh Phúc, local communities would previously enter the river as part of their daily lives. In Thái Bình, fear is instilled into younger members of the community by their elders to avoid 'deep holes' in the sand mining area and certain areas of the mangrove forest.

This highlights a significant change to livelihood activities linked to sand mining activity, whereby some areas of the mangrove have become dangerous, meaning members of all ages in the community can no longer access the land belonging to their village and commune. Although the sand mine is leased by sand mining companies who lease land temporarily, the mangrove forest affected by sand extraction supports livelihoods of farming community members, as seafood collecting supplements their daily earnings. Livelihoods are based on the mangrove, and its degradation directly undermines local land and resource access and perceived ownership (Bernett and Dearden, 2014). Both communities are dependent upon the river for resources and have used the river for leisure in the past. However, morphological change related to sand mining activities has directly impacted this, making communities cognisant of the negative impact of sand mining to their lives. The community are aware of alteration to the morphology of the Red River as their livelihoods have been directly impacted by anthropogenically-induced change, and especially sand mining (Marschke and Rousseau, 2022). This includes restricted access to their land and resources, loss of opportunities, reduced adaptive capacity and reduced willingness to partake in leisure activities like swimming and walking due to safety fears, invoking resistance in variable ways.

We argue that the proximity of threat and perceived illegality, transitioning from legal to illegal, hold the most influence upon protest behaviour to sand mining activities in the Vinh Phúc and Thái Bình villages (Fig. 3). Four key phase spaces are defined by the nature of protest behaviours exhibited by the village communities, ranging from cultural to structural and closed, hidden resistance (see ideas in Purnomo et al., 2021; Scott 1985; Scott 1990; Wicaksana and Amirudin,



**Fig. 3.** Conceptual diagram of resistance patterns in the Vinh Phúc and Thái Bình villages, based on Scott (1985) and Purnomo et al. (2021). Double-ended arrow indicates complementary resistance behaviours in time in Vinh Phúc, single-ended arrow indicates a permanent shift in behaviour in time in Thái Bình.

2021) within the community.

Where communities are in greater proximity to mining sites, they have greater perceived ownership of river-generated resources, and therefore enhanced threat to livelihoods. Villagers in Vĩnh Phúc utilise riverine sand and riverbanks for subsistence and semi-subsistence agriculture, and previously used the river for leisure activities and washing. The presence of sand mining and its impacts threatens those communities already living in precarious circumstances (see ideas in Conde and Le Billon, 2017); because of mining-induced livelihood impacts, conflict and protest occur in the village (see ideas in Bisht, 2021; Franks et al., 2014). The impacts of sand mining are more visible and direct, undermining housing and the dike built on the riverbank. Where proximity to and visibility of mining-derived impacts to livelihoods is greater, the strength of resistance is increased (Conde and Kallis, 2012). The scale of protest is also heightened by the perception of sand miners. Unlike in other cases, where sand miners can be perceived as “neighbours” (Wicaksana et al., 2021), people participating in resource extraction are perceived to be outside and external to both communities. The right to the resource is given to sand miners, rather than the local in-situ community, excluding the village from benefits of extraction (see ideas in Lamb et al., 2019). Because of the village’s cognition of both perceived right to the resource and impacts of sand mining, their resistance pattern is open and structural, where they are provided with official structural mechanisms, through reporting to the commune, district, and sea police via a 24/7 hotline, which is set out in legislation.

In particular, the allocation of resource rights to local people is cemented by the provincial government and legislation acknowledging sand mining is illegal in Vĩnh Phúc, thus enabling opportunities to complain to the authorities and openly protest. However, as the site of illegal mining is on the border of two provinces, this makes the regulation of sand mining difficult, resulting in distrust of local people to the effectiveness of reporting procedures, as observed in other areas (see ideas in Özkaynak and Rodriguez-Labajos, 2017). When communities engage in official structural protest mechanisms of reporting and petitioning, yet the presence of sand mining vessels is not reduced, the perceived proximity of the threat is heightened, leading to more open cultural forms of protest. This, alongside the threat of impact to livelihoods, the time-consuming nature of reporting to the commune late at night, and the inability to document evidence of illegal boats, leads to open cultural protest. Bankside protest, through making noise and throwing stones at vessels, and involvement of mass media, occur as the bankside community are desperate to protect their homes and livelihoods. However, continual protest is not always accessible, particularly where extraction occurs every day and night; significant time and resources are needed to protest sand mining (see ideas in Bebbington, 2010).

In contrast, where mining is legalised and ownership is granted to mining companies, the ability to openly protest becomes much diminished. Although communities are given a mechanism to complain about mining, the same response is given - that the mining site is legal, already in-place, with no solution for local people. This, combined with the relatively distal level of threat, prevents the community from complaining or openly protesting the mining operation to the commune. In these instances, community action transitioned from open, structural protest through reporting to commune authorities, to closed cultural protest, operating as within-village whisper or gossip. Where governments crack-down on larger-scale activism and protest, resistance becomes prohibited (Len, 2017). Resultantly, the community may be fearful of protesting openly. As the complaint cannot go further than commune-level, higher-up authorities at district and provincial level are not informed of community resistance to the sand mining, and so resistance operates in a relatively hidden fashion. In rural periphery regions, where extensive resource extraction has intensified relatively recently, communities utilise resistance structures available to them. This is a subtle form of resistance, occurring in everyday circumstances within the village as opposed to organised protest (Lassila, 2018;

Purnomo et al., 2021; Scott, 1985). Further, the spatial extent of the coastal village extends to the edge of the mangrove and does not contain the estuarine mining site. Despite many villagers utilising the mangrove and estuary for secondary livelihood and wellbeing activities, the land is not perceived to belong to the village. As a result, the village comply with the decision on legalised sand mining communicated by the commune authority.

#### 4. Conclusion

The demand for sand and gravel has led to accelerating sand mining throughout the Red River, where increases in the number of vessels, rapid development of mining equipment, and the extractive capacity of operations is resulting in the over-extraction of building materials from the Red River and its delta. This directly impacts the livelihoods and land belonging to local people through bank erosion, riverbed deepening, and alteration to riverine and coastal currents, particularly for those living near the river and mining sites. Within this resource conflict, resource rights are allocated to mining companies external to the village, leaving few, if any, benefits of resource extraction to local communities. Typically, the response of both the riverine and the coastal communities to resource exploitation is to protest existing power relations in a variety of ways. Communities exhibit their resistance through historically-routed cultural and structural mechanisms (Conde and Le Billon, 2017; Purnomo et al., 2021). Irrespective of whether sand mining is legalised or the proximity of mining sites, in both cases of Vĩnh Phúc and Thái Bình, community members share a common perspective - that sand mining directly impedes on their livelihoods and resource rights, resulting in unequal distribution of power across users and dependents of the resource.

Communities do not always engage openly in social movements, activism, and protest; we argue this is due to several factors including lack of time and resources and legislative constraints. Involvement in persistent protest and resistance reduces the ability of communities to engage in primary and secondary livelihood activities. This is not to say that communities accept the conditions of resource extraction, but have little choice to make their voices heard, especially with extrinsic political legislative barriers in place. Whilst there are no laws on protest in Vietnam, communities may still feel afraid to protest openly. At the same time, communities have a perceived ownership of the resource, both through culturally-routed recreational usage of the river and for livelihood support. This conflicts with official, structural mechanisms that remove the access rights of local people, instead granting resources and ownership to mining companies. This is heightened by the lack of benefits allocated to local communities, with all power given to mining companies. Where no compensation is given to communities in situ of mining sites, power imbalances such as these will continue to have a detriment impact on the livelihoods of local people.

The inability to openly protest keeps resistance hidden, highlighting the importance of micro-level studies with individual communities to better understand local people’s experiences and opinions of livelihood pressures (see ideas in Lamb et al., 2019). For the Red River Delta, and field of political ecology more broadly, a greater understanding of power dynamics and their influence on the rights of in-situ communities to equitable access and appreciation of who benefits from sand mining is of the upmost importance in order to understand resulting resistance dynamics. Further research is needed on characterising the impact and response of communities in sand mining regions, as well as considering the livelihoods of riverine and coastal sand miners, who we were not able to include in this study. In particular, using a political ecology lens, we encourage future studies of local-scale impacts to incorporate both internal socio-economic dynamics and characteristics of communities to gain a comprehensive understanding of how communities are impacted by and respond to resource extraction. Involving local, indigenous knowledge in the research process is critical where communities, unlike in other contexts, do not protest or openly express losses of resource



conflicts. Local in-situ communities have a right to resource allocation, experience impacts of resource over-extraction first hand, yet are not always involved in local decision-making processes (Rasch and Köhne, 2016). There is a need for advocacy groups and lobbyists to campaign for local village impacts and rights at and near sites of mining, as well as open spaces for communities to participate and have their voices heard by legislators.

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